

$\mathcal{K}_0^{\#1} \dagger$	$\mathcal{K}_0^{\#1} \alpha\beta\chi$							
$\mathcal{K}_0^{\#1} \dagger$	0	0						
$\mathcal{K}_0^{\#1} \dagger \alpha\beta\chi$	0	$3Y_1 k^2$	$\mathcal{K}_1^{\#1} \alpha\beta$	$\mathcal{K}_1^{\#2} \alpha\beta$	$\mathcal{K}_1^{\#1} \alpha$	$\mathcal{K}_1^{\#2} \alpha$		
$\mathcal{K}_1^{\#1} \dagger \alpha\beta$		0	0	0	0			
$\mathcal{K}_1^{\#2} \dagger \alpha\beta$		0	0	0	0			
$\mathcal{K}_1^{\#1} \dagger \alpha$		0	0	0	0			
$\mathcal{K}_1^{\#2} \dagger \alpha$		0	0	0	0			
						$\mathcal{K}_2^{\#1} \alpha\beta$	$\mathcal{K}_2^{\#1} \alpha\beta\chi$	
$\mathcal{K}_2^{\#1} \dagger \alpha\beta$						$-3k^2 \overset{(4)}{\kappa}_1$	0	
$\mathcal{K}_2^{\#1} \dagger \alpha\beta\chi$						0	$-3k^2 \overset{(4)}{\kappa}_1$	

	$\mathcal{T}_{0^+}^{\#1}$	$\mathcal{T}_{0^-}^{\#1}{}_{\alpha\beta\chi}$				
$\mathcal{T}_{0^+}^{\#1} \dagger$	0	0				
$\mathcal{T}_{0^-}^{\#1} \dagger^{\alpha\beta\chi}$	0	$\frac{1}{\text{Det}(0^-)}$	$\mathcal{T}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{T}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{T}_{1^-}^{\#1}{}_{\alpha}$	$\mathcal{T}_{1^-}^{\#2}{}_{\alpha}$
$\mathcal{T}_{1^+}^{\#1} \dagger^{\alpha\beta}$	0	0	0	0	0	0
$\mathcal{T}_{1^+}^{\#2} \dagger^{\alpha\beta}$	0	0	0	0	0	0
$\mathcal{T}_{1^-}^{\#1} \dagger^{\alpha}$	0	0	0	0	0	0
$\mathcal{T}_{1^-}^{\#2} \dagger^{\alpha}$	0	0	0	0	0	0
					$\mathcal{T}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{T}_{2^-}^{\#1}{}_{\alpha\beta\chi}$
$\mathcal{T}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$-\frac{1}{3k^2\kappa_1^{(4)}}$				0	
$\mathcal{T}_{2^-}^{\#1} \dagger^{\alpha\beta\chi}$	0				$-\frac{1}{3k^2\kappa_1^{(4)}}$	

Abbreviations used in matrices

$$Y_1 = 2 \overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_{15} \quad \&\& \quad \text{Det}(0^-) = 3 k^2 (2 \overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_{15})$$

Lagrangian

$$\begin{aligned}
& \overset{(4)}{\kappa}_1 \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta - 3 \overset{(4)}{\kappa}_1 \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta + 3 \overset{(4)}{\kappa}_1 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^\delta + \\
& 2 \overset{(4)}{\kappa}_{15} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^\delta - 12 \overset{(4)}{\kappa}_1 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta - 4 \overset{(4)}{\kappa}_{15} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta - \\
& 3 \overset{(4)}{\kappa}_1 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta - 2 \overset{(4)}{\kappa}_{15} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta + 6 \overset{(4)}{\kappa}_1 \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta \partial_\delta \mathcal{K}_{\chi\beta}{}^\delta - \\
& \overset{(4)}{\kappa}_{15} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta + \overset{(4)}{\kappa}_{15} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} - 6 \overset{(4)}{\kappa}_1 \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} - 2 \overset{(4)}{\kappa}_{15} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi}
\end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_0^{\#1} = 0$	1	$\partial_\beta \mathcal{J}^\alpha_{\alpha}{}^\beta = 0$	
$\mathcal{J}_1^{\#2\alpha} = 0$	3	$\partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi} = 0$	
$\mathcal{J}_1^{\#1\alpha} = 0$	3	$\partial_\chi \partial^\alpha \mathcal{J}^\beta_{\beta}{}^\chi + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha}_{\beta} = \partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi}$	
$\mathcal{J}_1^{\#2\alpha\beta} = 0$	3	$\partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\chi\alpha\beta} = \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta}$	
$\mathcal{J}_1^{\#1\alpha\beta} = 0$	3	$\partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi} = \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi}$	
Total # constraint(s):	13		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$-\frac{1}{\overset{(4)}{\kappa}_1}$
	2	0	$\frac{1}{\overset{(4)}{\kappa}_1}$
Resolved unitarity condition(s):		(Demonstrably impossible)	